

Melody Kuoch

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EDUCATION

University of Colorado Boulder — College of Engineering & Applied Science

Expected May 2029

B.A. Computer Science, Minor in Electrical Engineering

- Relevant coursework: Data Structures, Computer Systems, Circuits, Digital Logic, Calculus, Linear Algebra

TECHNICAL SKILLS

Languages: C, C++, Python, Java, SQL, Bash, HTML/CSS, JavaScript

Hardware & Embedded: Arduino, microcontroller programming, circuit prototyping, sensors, soldering

CAD & Fabrication: Onshape, SolidWorks, 3D printing (FDM), 3D scanning (Artec Leo), laser cutting, hardware troubleshooting & repair

Other: Git/GitHub, Figma, Agile, machine learning basics, technical documentation, teaching, adaptability

ENGINEERING EXPERIENCE

Engineering Student Support — CU Boulder Integrated Teaching & Learning Lab (ITLL) Aug 2025 – Present

University makerspace serving thousands of engineering students

- Maintain, calibrate, and repair 20+ FDM 3D printers, sustaining lab uptime across 39,000+ student prints and 109,000+ machine hours.
- Lead hands-on workshops teaching Arduino, CAD modeling, 3D printing, and 3D scanning to undergraduate and graduate engineering students.
- Mentor student engineers through the full project lifecycle — concept, CAD design, fabrication, testing, and troubleshooting.
- Wipe and re-image 200+ computers across the ITLL and Idea Forge to keep engineering workstations operational.
- Write technical documentation for reference by engineering support staff.

Engineering Instructor (Summer Program) — CU Boulder College of Engineering

Summer 2025

- Taught rising high school seniors the fundamentals of Arduino, CAD, laser cutting, and 3D printing through a hands-on build program.
- Guided teams from initial design concept to a working fabricated prototype.

PROJECTS

Automated Recycling Sorter

- Designing a hardware system that classifies and sorts recyclables using sensors, a microcontroller, and a machine learning model.
- Integrating CAD-designed, 3D-printed mechanical components with embedded control code.

Personal Portfolio Website

- Built and deployed a responsive personal site (HTML/CSS/JS) hosted on GitHub Pages to showcase projects and resume.

ADDITIONAL EXPERIENCE

Guest Service Representative — CU Boulder Recreation Center

Aug 2024 – May 2025

- Delivered front-line service in a high-volume campus facility while balancing a full-time engineering course load.

Food Service Worker — Voodoo Doughnut

Dec 2023 – Aug 2024

- Managed register operations and fast-paced production while maintaining full-time student status (20+ hrs/week).

LEADERSHIP

Lead Representative — Challenge 5280

Aug 2020 – Aug 2023

- Built weekly lesson plans and collected data/interviews to propose policy solutions across Denver Public Schools districts.